

A Technical Introduction to Instruction tuning

Section 1

Selection bias Selection bias refers to the inherent tendency of large language models to favor certain option identifiers irrespective of the actual content of the options. This bias primarily stems from token bias—that is, the model assigns a higher a priori probability to specific answer tokens (such as "A") when generating responses. As a result, when the ordering of options is altered (for example, by systematically moving the correct answer to different positions), the model's performance can fluctuate significantly. This phenomenon undermines the reliability of large language models in multiple-choice settings.

Section 2

An LLM is a type of foundation model (large X model) trained on language. LLMs can be trained in different ways. In particular, GPT models are first pretrained to predict the next word on a large amount of data, before being fine-tuned.

Section 3

As an example, consider a tokenizer based on byte-pair encoding. In the first step, all unique characters (including blanks and punctuation marks) are treated as an initial set of n-grams (i.e. initial set of uni-grams). Successively the most frequent pair of adjacent characters is merged into a bi-gram and all instances of the pair are replaced by it. All occurrences of adjacent pairs of (previously merged) n-grams that most frequently occur together are then again merged into even lengthier n-gram, until a vocabulary of prescribed size is obtained. After a tokenizer is trained, any text can be tokenized by it, as long as it does not contain characters not appearing in the initial-set of uni-grams.

Section 4

Adversarial evaluations LLMs' rapid improvement regularly renders benchmarks obsolete, with the models exceeding the performance of human annotators. In addition, "shortcut learning" allows AIs to "cheat" on multiple-choice tests by using statistical correlations in superficial test question wording to guess the correct responses, without considering the specific question. Some datasets are adversarial, focusing on problems that confound LLMs. One example is the TruthfulQA dataset, a question answering dataset consisting of 817 questions that stump LLMs by mimicking falsehoods to which they were exposed during training. For example, an LLM may answer "No" to the question "Can you teach an old dog new tricks?" because of its exposure to the English idiom you can't teach an old dog new tricks, even though this is not literally true. Another example of an adversarial evaluation dataset

is Swag and its successor, HellaSwag, collections of problems in which one of multiple options must be selected to complete a text passage. The incorrect completions were generated by sampling from a language model. The resulting problems are trivial for humans but defeated LLMs. Sample questions:

Section 5

Training of largest language models might need more linguistic data than naturally available, or that the naturally occurring data is of insufficient quality. In these cases, synthetic data might be used.

Section 6

Moving beyond n-gram models, researchers started in 2000 to use neural networks as language models. Following the breakthrough of deep neural networks in image classification around 2012, similar architectures were adapted for language tasks. This shift was marked by the development of word embeddings (e.g., Word2Vec by Mikolov in 2013) and sequence-to-sequence (seq2seq) models using LSTM. In 2016, Google transitioned its translation service to neural machine translation (NMT), replacing statistical phrase-based models with deep recurrent neural networks. These early NMT systems used LSTM-based encoder-decoder architectures, as they preceded the invention of transformers. At the 2017 NeurIPS conference, Google researchers introduced the transformer architecture in their landmark paper "Attention Is All You Need". This paper's goal was to improve upon 2014 seq2seq technology, and was based mainly on the attention mechanism developed by Bahdanau et al. in 2014. The following year in 2018, BERT was introduced and quickly became "ubiquitous". Though the original transformer has both encoder and decoder blocks, BERT is an encoder-only model. Academic and research usage of BERT began to decline in 2023, following rapid improvements in the abilities of decoder-only models (such as GPT) to solve tasks via prompting. Although decoder-only GPT-1 was introduced in 2018, it was GPT-2 in 2019 that caught widespread attention because OpenAI claimed to have initially deemed it too powerful to release publicly, out of fear of malicious use. GPT-3 in 2020 went a step further and as of 2025 is available only via API with no offering of downloading the model to execute locally. But it was the 2022 consumer-facing chatbot ChatGPT that received extensive media coverage and public attention. The 2023 GPT-4 was praised for its increased accuracy and as a "holy grail" for its multimodal capabilities. OpenAI did not reveal the high-level architecture and the number of parameters of GPT-4. The release of ChatGPT led to an uptick in LLM usage across several research subfields of computer science, including robotics, software engineering, and societal impact work. In 2024, OpenAI released the reasoning model OpenAI o1, which generates long chains of thought before returning a final answer. Many LLMs with

parameter counts comparable to those of OpenAI's GPT series have been developed. Since 2022, weights-available models have been gaining popularity, especially at first with BLOOM and LLaMA, though both have restrictions on usage and deployment. Mistral AI's open-weight models Mistral 7B and Mixtral 8x7B have a more permissive Apache License. In January 2025, DeepSeek released DeepSeek R1, a 671-billion-parameter open-weight model that performs comparably to OpenAI o1 but at a much lower price per token for users. Since 2023, many LLMs have been trained to be multimodal, having the ability to also process or generate other types of data, such as images, audio, or 3D meshes. Open-weight LLMs have become more influential since 2023. Per Vake et al. (2025), community-driven contributions to open-weight models improve their efficiency and performance via collaborative platforms such as Hugging Face.

Section 7

In the context of training LLMs, datasets are typically cleaned by removing low-quality, duplicated, or toxic data. Cleaned datasets can increase training efficiency and lead to improved downstream performance. A trained LLM can be used to clean datasets for training a further LLM. With the increasing proportion of LLM-generated content on the web, data cleaning in the future may include filtering out such content. LLM-generated content can pose a problem if the content is similar to human text (making filtering difficult) but of lower quality (degrading performance of models trained on it).